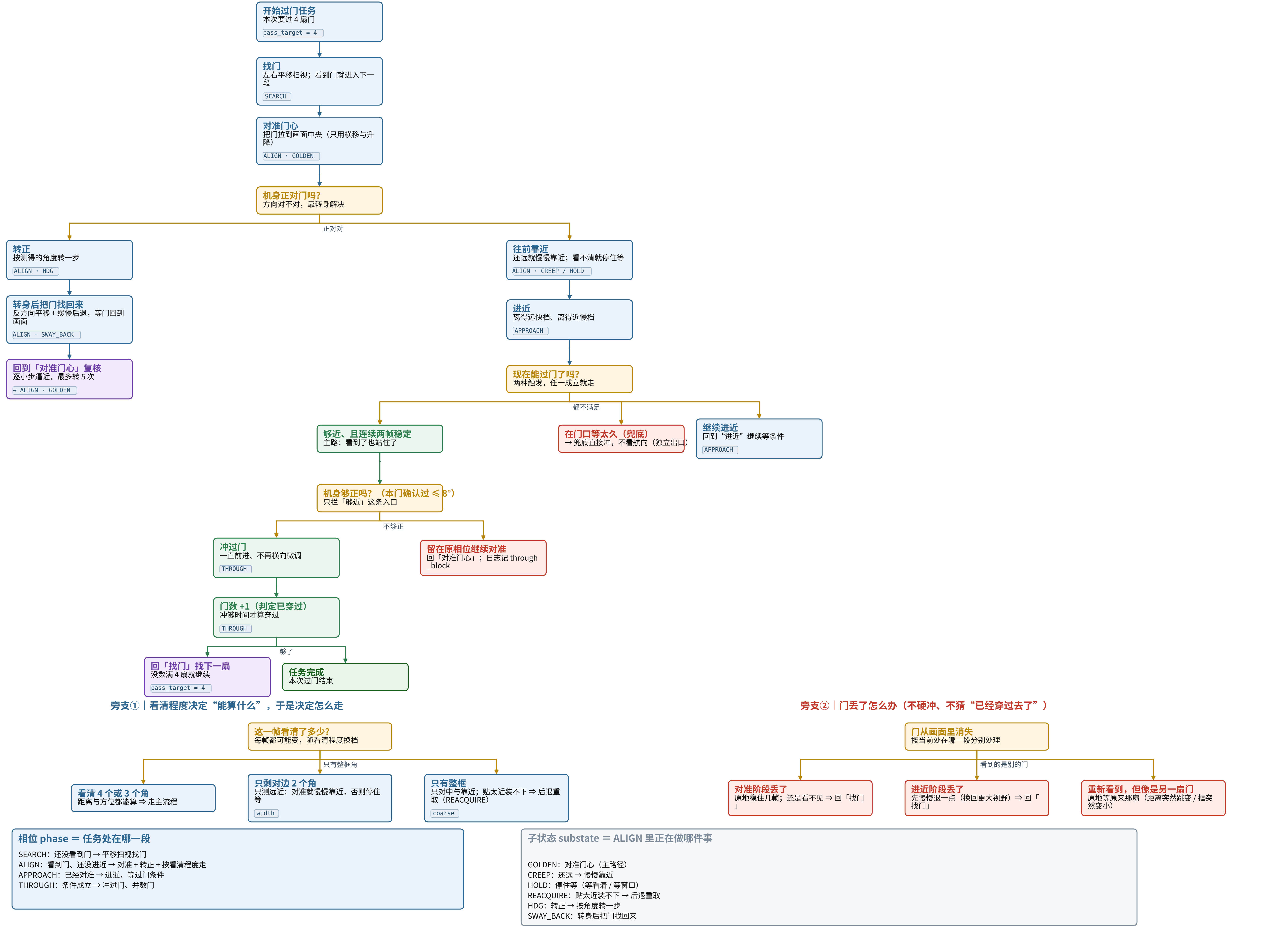


过门流程图（设计层面）

从“找门”到“过门完成”：每一段要做什么、什么时候转下一段；小标签是约定名称（相位 / 子状态），数值判据在最底部



数值判据速查 · 直接取自 cfg/comm.yaml（gate: 段，键名省略 comm.gate. 前缀）与 cfg/vision.yaml

够近才冲	cross = 1.2 · cross_confirm_frames = 2
够正才冲	require_align_deg = 8.0
对准稳定	px_x = 0.20 · px_y = 0.25 · confirm_frames = 2
门口兜底	enable = true · near_ratio = 0.60 · dx_max = 0.14 · dy_max = 0.20 · timeout_ms = 3000
丢门处置	pose_hold_frames = 10 · lost_backward = 0.20
后退重取	max_ms = 800 · stop_ratio = 0.75 · max_times = 2 · reset_after_ms = 3000
转正	tol_deg = 8.0 · turn_scale = 0.8 · max_step_deg = 15.0 · max_turns = 5 · turn_timeout_s = 8.0

转身回找
前进速度
进近分档
找门扫视
换档门档
任务收尾
安全

post_sway = 0.20 · post_sway_back = 0.15 · post_sway_ms = 600.0 · post_sway_kpt_min = 2
creep = 0.20 · reacquire = 0.25 · through = 0.6
surge_fast = 0.35 · surge_slow = 0.15 · slow_max = 1.3
sway = 0.6 · sweep_s = 2.0 · sweep_max_s = 32.0 · pause_s = 1.0
z_max = 1.5 · far_ratio = 0.18 · near_ratio = 0.55
pass_target = 4 · timeout_ms = 180000
min_depth_m = 0.55 · max_z_jump_m = 0.8